

ADR 0004 — Keeper-star actionable classification for near-duplicate groups

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Context

Near-duplicate groups are the connected components of the verified within-threshold pair graph: MIH finds every pair within hamming distance `--threshold` (recall proven against a brute-force scan), and union-find merges chains of such pairs into one group. Transitivity is the point — a burst of edits $A \rightarrow B \rightarrow C$ belongs together for *review* — but it silently implies a guarantee nobody measured: A and C can sit in one group while their pairwise distance exceeds the threshold. Every non-keeper member was counted as a reclaimable duplicate regardless, and `hamming_to_keep` was display-only. The v1.2 contract already commits to "near duplicates are never selected automatically"; v1.1's plan generation needs a representation that makes the safe subset explicit, per file, before any action code exists. That is issue #9's mandate, and the future-roadmap design named the two candidate shapes: "explicit edges or keeper-star-safe groups."

Decision

- **Review groups stay exactly as they are.** Union-find components remain the group shape — no re-partitioning, no group splits. Their honest meaning ("connected by a chain of verified pairwise matches") is now documented on `near_components`, the extracted grouping function.
- **Keeper-star classification, per member, additive.** A member is `actionable` only when its distance to the keeper was *directly measured* and is within the run's threshold — exact members by content identity, near members by `hamming_to_keep <= threshold`; keepers, shadowed rows and hardlinks never. One pure function (`member_is_actionable`, fail-closed) is the single source of the rule, echoed in-band as `params.actionable_rule`.
- **Accounting counts only what the rule admits.** `reclaimable_bytes` and `duplicate_files` now cover actionable members only; transitive-only members land in new additive counters (`Group.review_only_bytes`, `Summary.transitive_only_files / review_only_bytes`) and carry a `[review-only]` tag in the terminal rendering. This is a value-level correctness fix to match the fields' documented meaning, shipped inside report `version: 1` under the additive-only policy; the fixture re-bless diff shows only added keys and unchanged existing values.
- **The oracle is the acceptance bar.** A grouping-level brute-force oracle (`grouping_matches_brute_force_oracle_at_every_threshold`) proves at every threshold $0..=3$ that pipeline components equal the components of the brute-force pair graph, and that classification admits exactly the members measured within threshold of *any* keeper choice — with an anti-vacuity assertion that the corpus really produces transitive-only members. A real-

image chain (A 640×480 keeper, B at distance 2, C at distance 4 via B) exercises the same guarantee end-to-end and is frozen non-degenerately in

`tests/fixtures/contract/dedup_chain.json`.

Alternatives considered

- **Explicit pairwise edge lists in the JSON** — the other shape the roadmap design named. Rejected: $O(n^2)$ payload on large groups, overspecifies what v1.1 planning needs, and the safety question a plan must answer is only ever "is this file measured-safe relative to the keeper being retained" — exactly the star, not the full graph. Edges can be added later additively if a UI wants them.
- **Re-partitioning clusters so every group is pairwise-within-threshold** (e.g. splitting components until diameter $\leq$ threshold) — rejected: destroys the review affordance (edit chains scatter across groups), partition choice is unstable under member insertion, and it still needs a keeper rule afterwards; safety comes from classification, not from reshaping groups.
- **Client-side derivation (`hamming_to_keep` vs `params.threshold`) instead of a field** — rejected: every consumer would re-implement the rule, including its shadowed/hardlink/keeper exclusions, and one of them would get it wrong; the fail-closed rule belongs in exactly one place.
- **Overloading exit code 2 or `has_skips()` to flag transitive-only members** — rejected: exit 2 is documented as archive reachability/completeness; a routine, correct grouping outcome is not a skip condition.

Consequences

- v1.1 plan generation gets its hard selection floor for free: only `actionable: true` members may ever be auto-proposed, and the v1.2 "explicit per-file selection" contract for near dups has a typed, fixture-frozen representation to hang selections on.
- `max_distance` keeps its documented keeper-relative meaning and now has company: a group whose `max_distance` exceeds the threshold visibly carries review-only members instead of silently inflating reclaimable space. Group ordering also gained a deterministic tie-break (smallest member identity), closing the documented HashMap-order fragility noted in the contract corpus.
- Every dedup fixture re-bless in this change is additive-only; `dedup_chain.json` joins the frozen set (8 fixtures) so the new fields are pinned at non-degenerate values per ADR 0003's residual-gaps note.
- Reported reclaimable numbers shrink on corpora with transitive chains — by exactly the bytes that were never measured safe. README and compatibility docs state the fix; scripts consuming `reclaimable_bytes` see honest values, not a shape change.
- The oracle pattern extends: v1.3's benchmarked MIH work (thresholds above 3) inherits a grouping-level oracle it must keep green, not just the pairwise recall test.